

Ian Kim

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EDUCATION

Johns Hopkins University

MSE in Computer Science

Baltimore, MD

December 2027

- **Coursework:** Computer Vision, NLP: Self-Supervised, Computer Networks

Williams College

Williamstown, MA

BA in Computer Science, *GPA: 3.61/4.0*

June 2026

- **Coursework:** Computer Security, Foundations of Artificial Intelligence, Algorithm Design and Analysis, Data Structures and Advanced Programming, Computational Analysis of Big Data, Computer Organization, Discrete Mathematics

TECHNICAL SKILLS

Programming Languages: C, Python, SQL, R, C#, Java, JavaScript/TypeScript, HTML/CSS

Tools/Frameworks: React, Node.js, PostgreSQL, MongoDB, Git, Redis, Capacitor, Supabase, Unity Game Engine

WORK EXPERIENCE

Johns Hopkins S4 Lab, Research Assistant

September 2026 - Present

Baltimore, MD

- Building systems and networking infrastructure for a LiDAR-based smart intersection

WorldCare, Artificial Intelligence Research Intern

January 2026

Remote

- Developed clinical data de-identification pipeline to process protected health information while ensuring HIPAA compliance; designed modular architecture using Python, Presidio, and Faker
- Engineered context-aware anonymization module and clinical filtering system to intelligently replace personally identifiable information while distinguishing medical terminology from protected health information
- Conducted research analyzing privacy and security risks of large language model deployment with protected health information; collaborated with two-person team to document PHI exposure vulnerabilities and present AI safety mitigation strategies to technical leadership

Professor James Bern Lab Group, Research Assistant

June 2025 – Aug 2025

Williamstown, MA

- Built a real-time interaction pipeline using Unity, Teensy microcontroller, and ODrive motor controller to simulate tactile object collisions with robotic haptic feedback
- Developed a C-based driver program to coordinate VR object tracking, collision detection, and motor control in real time
- Conducted controlled experiments measuring haptic feedback accuracy and motor response latency; collaborated with a four-person team to refine the mixed reality system

Williams College, Teaching Assistant - Data Structures & Algorithms

September 2025 – June 2026

Williamstown, MA

- Led weekly labs covering core data structures, algorithms, and computational complexity
- Guided students through debugging, algorithm optimization, and reasoning about performance tradeoffs
- Reviewed student code and provided structured feedback focused on correctness, clarity, and efficiency

PROJECTS

PokéNet, Personal Project

August 2026

- Built a fine-grained image recognition system for 151 Pokémon species using deep metric learning, embedding images into a 128-D similarity space via a frozen OpenCLIP backbone with a trained projection head
- Fixed embedding collapse in triplet loss training by redesigning the mining strategy and loss function, unlocking stable convergence
- Hit 90% top-1 accuracy, an 18-point jump over the zero-shot CLIP baseline, using a k-means clustered reference gallery for nearest-neighbor lookup

RedPrompt, Computer Security Final Project

May 2026

- Engineered a local AI application environment using Ollama and Streamlit to simulate a realistic deployment target for prompt injection attack research
- Developed a formal testing framework to systematically evaluate prompt injection efficacy against a locally hosted DeepSeek model, measuring attack success rates across multiple injection strategies

LEADERSHIP & COMMUNITY ENGAGEMENT

Williams Varsity Men's Lacrosse, Player

September 2022 - June 2026

Berkshire Center for International Policy, Member

September 2025 - June 2026